

WAN, WMS & WAC SERIES (2 TO 100+ TR)

Specialized Application Process Chillers

Custom Titanium Anodizing Chillers, N+1 Redundant Medical Scan Chillers & Severe Chemical Acid Cooling Systems



WAN SERIES

Pure Titanium Gr. 2
Anodizing Chillers (2-
100+ TR)

WMS SERIES

N+1 Redundant MRI/CT
Chillers with City Auto-
Bypass

WAC SERIES

Hastelloy / PTFE Severe
Acid Process Chillers

ZERO DOWNTIME

Fail-Safe Engineering for
Critical Life & Asset Care

THERMODYNAMIC ARCHITECTURE & ENGINEERING

Operational Overview & Engineering Advantages

Continuous industrial duty engineered for 45°C tropical ambient conditions

Standard commercial chillers fail rapidly when exposed to aggressive acid electrolytes or when supporting zero-downtime medical imaging magnets. Win Equipments specializes in custom-engineered thermal management packages: WAN Series Titanium Anodizing Chillers resisting 20% sulfuric acid baths; WMS Series Medical Scan Chillers providing dual-redundant cooling for clinical MRI/CT magnets; and WAC Series Acid Process Chillers featuring Hastelloy C-276 metallurgy and sealless magnetic drive pumps.

Working Principle: WAN Anodizing Chillers utilize Grade 2 pure Titanium seamless immersion coils or external Titanium plate heat exchangers to absorb intense exothermic rectifier heat, holding bath temperature at +18°C to +21°C for Type II anodizing or -2°C to +4°C for Type III hardcoat anodizing. WMS Medical Chillers feature dual independent refrigeration circuits (2 × 100% capacity) and dual SS316 pumps with auto-failover and an emergency motorized city water bypass that engages automatically upon electrical power failure to prevent catastrophic liquid helium boil-off in superconducting MRI magnets.

SYSTEM INTEGRATION & P&ID FLOW ARCHITECTURE

Severe Chemical Bath / MRI Cold Head → Acid-Resistant Pumping Circuit → Grade 2 Titanium / Hastelloy Exchanger → Redundant Modulated Chillers → Closed Return



Grade 2 Pure Titanium Acid Immersion Metallurgy

Features seamless Grade 2 Titanium heat exchanger coils fabricated with inert gas TIG welding. Completely impervious to 20% sulfuric, chromic, and oxalic acid anodizing solutions, eliminating catastrophic acid leaks into refrigeration circuits.



Dual Redundant (N+1) Circuits & Auto City Bypass

WMS Series incorporates two completely separate refrigeration compressors, independent condensers, and dual SS316 pumps. Upon facility power interruption, a spring-return motorized valve opens municipal water to cold heads, preventing ₹30L helium boil-off.



Hastelloy C-276 & PTFE Fluoropolymer Options

For highly aggressive pickling acids (hydrochloric, nitric, hydrofluoric), WAC chillers deploy exotic Hastelloy C-276 or solid PTFE tube bundles coupled with sealless PVDF magnetic drive pumps with zero dynamic mechanical seals.



Ultra-Quiet Hospital & Lab Acoustics

WMS medical chillers utilize acoustically insulated compressor enclosures and low-RPM sickle-blade condenser fans, reducing ambient noise to under 65 dBA for hospital courtyard and rooftop installation.



CERTIFIED ENGINEERING RATINGS

Technical Specifications Matrix

Full standard model range & operational parameters

Standard Design Conditions: WAN Anodizing: Bath +18°C to +21°C (Hardcoat -2°C to +4°C) | WMS Medical: Supply +8°C to +15°C (±0.2°C) | WAC Acid: Acid bath -5°C to +15°C | Exchanger: Titanium Gr. 2 / Hastelloy / PTFE

MODEL CODE	SERIES / APP	CAPACITY TR	COOLING KW	RECTIFIER / DUTY	WETTED METALLURGY	COMPRESSOR SETUP	FLOW LPM
WAN 020	WAN Anodizing	2.0 TR	7.0 kW	Up to 350 A	Pure Titanium Gr. 2	Copeland Scroll	25 LPM
WAN 050	WAN Anodizing	5.0 TR	17.5 kW	Up to 1,000 A	Pure Titanium Gr. 2	Copeland Scroll	55 LPM
WAN 100	WAN Anodizing	10.0 TR	35.1 kW	Up to 2,000 A	Pure Titanium Gr. 2	Copeland / Danfoss	110 LPM
WAN 200	WAN Anodizing	20.0 TR	70.3 kW	Up to 4,000 A	Pure Titanium Gr. 2	Dual Scroll Circuits	215 LPM
WAN 500	WAN Anodizing	50.0 TR	175.8 kW	Up to 10,000 A	Pure Titanium Shell & Tube	Bitzer Semi-Hermetic	530 LPM
WAN 1000	WAN Anodizing	100.0 TR	351.6 kW	Up to 20,000 A	Pure Titanium Shell & Tube	Twin Semi-Herm / Screw	1,050 LPM
WMS 030	WMS Medical	3.0 TR	10.5 kW	Clinical MRI / CT	SS316L / Copper	Dual 2×1.5 TR Redundant	32 LPM
WMS 050	WMS Medical	5.0 TR	17.5 kW	1.5T / 3.0T MRI	SS316L / Copper	Dual 2×2.5 TR Redundant	52 LPM
WMS 100	WMS Medical	10.0 TR	35.1 kW	High-Field Scanner	SS316L / Copper	Dual 2×5.0 TR Redundant	105 LPM
WMS 200	WMS Medical	20.0 TR	70.3 kW	Dual Scanner Suite	SS316L / Copper	Dual 2×10.0 TR Redundant	210 LPM
WAC 050	WAC Acid Chiller	5.0 TR	17.5 kW	Pickling / Battery	Hastelloy / PTFE	Copeland Scroll	50 LPM
WAC 150	WAC Acid Chiller	15.0 TR	52.7 kW	Exothermic Reactor	Hastelloy / PTFE	Bitzer Semi-Hermetic	150 LPM

ENGINEERING SELECTION & OPERATIONAL NOTES

- Rectifier Sizing Formula for Anodizing: Required TR = (Volts × Amps × 3.412) ÷ 12,000 × 1.15 safety factor.
- Emergency City Water Auto-Bypass valve included as standard on all WMS Medical Scan units.
- Positive-pressure nitrogen purged NEMA 4X electrical enclosures available for severe acid pickling plant environments.

FIELD DEPLOYMENT & COMPETITIVE ANALYSIS

Applications & Industrial Benchmarks

Demonstrated operational benefits across high-reliability manufacturing sectors

Architectural & Aerospace Anodizing

Maintains bath temperatures within $\pm 0.5^{\circ}\text{C}$ during high-amperage cycles, ensuring uniform oxide thickness and vibrant dye uptake.

Hospital Diagnostic Imaging (MRI / CT / PET)

Protects superconducting magnets from cryogenic helium boil-off with dual 100% redundant refrigeration and emergency water failover.

Lead-Acid & Lithium Battery Formation

Dissipates intense exothermic heat generated during high-rate battery charging, preventing plate bucking and thermal runaway.

Chemical Steel Pickling & Passivation

Cools aggressive hydrochloric and nitric acid baths without corrosion, extending acid bath life and ensuring uniform etching.

Spot Chilling & Precision Optical Grinding

Delivers pinpoint micro-fluid chilling directly to cutting tools and optical glass polishing chucks to eliminate thermal drift.

Engineering Comparison: Win Specialized vs. Generic Package Chillers		Standard ISO 9001:2015 Quality
Engineering Parameter	Win Equipments Engineered Solution	Standard / Imported Benchmark
Wetted Metallurgy	Pure Titanium Grade 2 / Hastelloy C-276 / Solid PTFE	Copper or standard SS304 (Rapid acid pinhole failure)
Redundancy	True N+1 dual independent circuits + auto city water bypass	Single circuit; entire hospital imaging stops on trip
Pumping System	Dual SS316 or Sealless PVDF Magnetic Drive Pumps	Standard mechanical seal pumps prone to chemical leakage
Custom Sizing	Direct calculation based on rectifier wattage & bath kinetics	Generic rule-of-thumb resulting in severe undersizing

MANUFACTURING STANDARDS & FACTORY VERIFICATION

Factory Quality Assurance & Consultation

Calibrated testing regimes, global certifications, and direct engineering contact

✓ Factory Acceptance Testing (FAT)

- 100% Dye-penetrant and pneumatic leak testing on Titanium TIG weld joints
- Automated simulated failover test: Circuit 1 compressor trip → Circuit 2 engagement in <1 sec
- Emergency city water bypass motorized valve actuation and spring return verification
- Continuous 12-hour full-load thermal calorimeter test with simulated bath heat

★ Certified Standards & Accreditations

- ISO 9001:2015 Quality Management System
- IAF (International Accreditation Forum) Accredited
- DAC (Dubai Accreditation Center) Recognized
- Pre-dispatch run-testing under tropical temperature simulation



Manufacturer Warranty Assurance: 24 Months warranty on Titanium heat exchangers; 12 months comprehensive on system components. Direct field service engineering and genuine replacement spare parts stocked at Arasur Works, Coimbatore.

WIN EQUIPMENTS

SAVE WATER AND POWER • ISO 9001:2015 (IAF & DAC ACCREDITED)

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SCAN FOR WEB
SPECS